

Artificial Intelligence Fundamentals

MSc on AI / 2026-2027

David Chaves-Fraga

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Singular Research Center on
Intelligent technologies

Teaching Team - Theory

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David Chaves-Fraga (coordinator)

Office 105 - CiTIUS

<https://citius.gal/team/david-chaves-fraga/>





Teaching Team - Labs



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**David Chaves
Fraga**

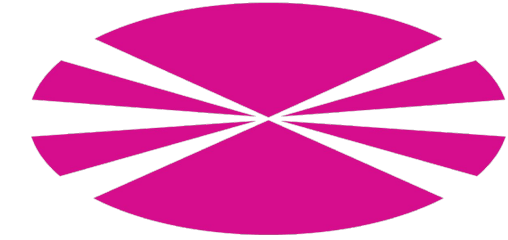


**Antonio Moscoso
Sanchez**

UniversidadeVigo



**David Candal
Ventureira**



**Alejandro Mayorga
Redondo**



**Daniel Llamas
Gómez**



School & Faculties - USC



Virtual USC▼ Electronic headquarters Gobierno▼ Centres Departments Services Personnel Directory Transparency▼ Xornal

USC Courses Admission Research USC Life

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Higher Technical Engineering School

The center▼ Studies▼ Academic information▼ Quality▼



Higher Technical Engineering School

Rúa Lope Gómez de Marzoa, s/n, 15782 Santiago de Compostela
881816700 (Conserxaría)
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The Higher Technical School of Engineering is centre of the University of Santiago de Compos responsible for the management and organisat studies in the field of engineering. The mission the ETSE is to contribute to comprehensive education, in a stimulating environment of innovation and human relations, of engineers

Fundamentos de IA

2024/2025

Información

Programa

Profesorado

Horarios de clases, exámenes e idiomas de uso

Créditos ECTS

Créditos ECTS: 3

Horas ECTS Criterios/Memorias

Horas de Tutorías: 1
Clase Expositiva: 10
Clase Interactiva: 11
Total: 22

Lenguas de uso

Inglés

Tipo:

Materia Ordinaria Máster RD 1393/2007 - 822/2021

Departamentos:

[Electrónica y Computación](#)

Áreas:

Ciencia de la Computación e Inteligencia Artificial

Centro

[Escuela Técnica Superior de Ingeniería](#)

Convocatoria:

Primer semestre

Docencia:

Con docencia

Matrícula:

Matriculable | 1ro curso (Si)



School & Faculties - UDC

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facultade de
informática
da coruña

A FIC

ESTUDOS

CALIDADE

SERVIZOS

INTERNATIONAL

+ FIC

NOVAS / EMPREGO





Español

Galego

Horarios curso 2024/25 corr

Ligazón incorrecta aos horarios d

Curso 2024-25

Facultade de Informática da

Últimas Novas

Máster Universitario en Intelixencia Artificial

Asignaturas

Fundamentos de IA

galego

castellano

english



Datos Identificativos					2024/25			
Asignatura (*)	Fundamentos de IA			Código	614544001			
Titulación	Máster Universitario en Intelixencia Artificial							
Descriptores	Ciclo	Período	Curso	Tipo	Créditos			
	Mestrado Oficial	1º cuatrimestre	Primeiro	Obrigatoria	3			
Idioma	Inglés							
Modalidade docente	Presencial							
Prerrequisitos								
Departamento	Ciencias da Computación e Tecnoloxías da Información							
Coordinación	Mosqueira Rey, Eduardo	Correo electrónico	eduardo.mosqueira@udc.es					
Profesorado	Mosqueira Rey, Eduardo Munteanu , Cristian Robert	Correo electrónico	eduardo.mosqueira@udc.es c.munteanu@udc.es					
Web	http://www.usc.gal/es/estudios/masteres/ingenieria-arquitectura/master-universitario-intelixencia-artificial/20232024/fundament							
Descrición xeral	A materia introduce ao estudante nos aspectos básicos que definen a IA, fundamentalmente a resolución automática de problemas non abordables ou dificilmente abordables mediante técnicas convencionais de programación. Neste contexto, abordaranse os algoritmos de procura no espazo de estados para a resolución de problemas, así como a representación de coñecemento e o razoamento Guía docente centro coordinador (USC): https://www.usc.gal/es/estudios/masteres/ingenieria-arquitectura/master-universitario-intelixencia-artificial/20232024/fundamentos-ia-18827-17978-2-102307							
Competencias / Resultados do título		Resultados de aprendizaxe		Contidos				
Planificación		Metodoloxías		Atención personalizada				
Avaliación		Fontes de información		Recomendacións				



School & Faculties - UVigo

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Universidade de Vigo

CAMPUS AUGA BIBLIOTECA DEPORTES CULTURA CORREO UVIGO MOOVI DUVI

SECRETARÍA GL | ES





A ESEI Docencia Estudos I+D e empresas Calidade Actualidade



titica

Máster universitario en Intelixencia artificial

L Materias

L Fundamentos de IA

galego

castellano

english

DATOS IDENTIFICATIVOS

2024_25

Materia	Fundamentos de IA	Código	O06M193V01101	
Titulación	Máster universitario en Intelixencia artificial			
Descritores	Cr.totais	Sinale	Curso	Cuadrimestre
	3	OB	1	1c
Lingua de impartición				
Prerrequisitos				
Coordinador/a	García Lourenco, Analia María	Correo-e	analia@uvigo.es	
Profesorado	García Lourenco, Analia María			
Web	http://guiadocente.udc.es/guia_docent/index.php?centre=614&ensenyament=614544&consulta=assignatures&any_academic=2023_24			
Descrición xeral				
Resultados de Formación e Aprendizaxe		Resultados previstos na materia		Contidos
Planificación		Metodoloxía docente		Atención personalizada
Avaliación		Bibliografía. Fontes de información		Recomendacións

de Vigo

| Reitoría | Campus Universitario | C.P. 36.310 Vigo (Pontevedra) | España | Tlf: +34 986 812 000

Objectives

The course introduces the student **to the basic aspects that define AI**, mainly the **automatic resolution of problems** that are not or hardly approachable by conventional programming techniques.

In this context, **state space search algorithms for problem solving** will be addressed, as well as **knowledge representation and reasoning**.

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Learning outcomes

To know the **fundamental principles and basic techniques** of artificial intelligence.



Contents - Block I

1. Introduction (now)

2. Problem solving in AI

- Search algorithms
- Uninformed search algorithms
- Informed (heuristic) search strategies
- Heuristic functions
- Search for Constraint satisfaction problems

Contents - Block II

3. Structured representations of knowledge

- Introduction to rule-based systems:
 - Basic components
 - Inference
 - Conflicts resolution
- Introduction to ontologies

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4. Methods of knowledge representation

- Frames.
- Semantic networks.
- Knowledge graphs.

5. Basic models of reasoning

- First-order logic.
- Descriptive logic.
- SPARQL.

Bibliography

- Russell, S., Norvig, P. Artificial Intelligence (A Modern Approach), (4th Edition Global Edition, 2022). ISBN: 9781292401133.
- R. Marín, J.T. Palma, Inteligencia Artificial. Técnicas, Métodos y aplicaciones. McGraw-Hill, 2008. ISBN 978-84-481-5618-3.
- Fernández Galán, S., González Boticario, J., Mira Mira, J. Problemas resueltos de inteligencia artificial aplicada: búsqueda y representación. Addison Wesley (1998). ISBN: 9788478290178
- Gómez-Pérez, A., Fernández-López, M., Corcho, O. (2006). Ontological Engineering: with examples from the areas of Knowledge Management, e-Commerce and the Semantic Web. Springer Science & Business Media. Springer-Verlag. ISBN: 978-1-85233-551-9.
- C. Maria Keet, An introduction to ontology engineering. College Publications, 218. ISBN: 978-1848902954.

Complementary references

- Nilsson, N.J. Artificial Intelligence (A New Synthesis). McGraw-Hill (2001). ISBN: 9788448128241
- Virginie Mathivet. Inteligencia Artificial para desarrolladores. ENI Editions, 2015.
- Fernando Sancho Caparrini. Curso de Inteligencia Artificial. <http://www.cs.us.es/~fsancho>
- Handbook of Artificial Intelligence. Springer-Verlag, 2015. ISBN 978-3-662-43505-2.

Competences

Basic and general

- CG1 - Maintaining and extending grounded theoretical approaches to enable the introduction and exploitation of new and advanced technologies in the field of Artificial Intelligence.
- CG3 - Search and select the useful information needed to solve complex problems, handling with fluency the bibliographic sources of the field.
- CB6 - Possess and understand knowledge that provides a basis or opportunity for originality in the development and/or application of ideas, often in a research context.
- CB8 - Students are able to integrate knowledge and deal with the complexities of making judgements based on incomplete or limited information, including reflections on the social and ethical responsibilities associated with the application of their knowledge and judgements

Competences

Transversal

- CT2 - Master the oral and written expression and comprehension of a foreign language.
- CT3 - Use the basic tools of information and communication technologies (ICT) necessary for the exercise of their profession and for lifelong learning.
- CT4 - To develop for the exercise of a citizenship that respects democratic culture, human rights and the gender perspective.
- CT6 - Acquire life skills and healthy habits, routines and lifestyles.
- CT8 - To value the importance of research, innovation and technological development in the socio-economic and cultural progress of society.

Competences

Specific

- CE4 - To know the fundamentals and basic techniques of artificial intelligence and its practical application.

Teaching methodology - Theory

- MS Teams platform of the three universities as the basic platform
 - virtual tutoring of students
- [AI Master website](#)
- Teaching methodology based essentially on individual work for exercises and group work for interactive sessions
- Theoretical sessions
 - face to face at USC
 - streamed online for UdC and UVIGO students: <https://mia.udc.es/2026-27/>

Teaching Methodology - Labs

- Face2Face in the IT lab at each university
- Use of software tools
- Developing applications for each of the thematic blocks:
 - Groups of three people
 - No specific programming language recommendation: group decision

Assessment System

- Theory: 50%
 - Final exam
 - Minimum threshold: 4/10
- Labs: 50%
 - Two deliverables, same weight (50% each)
 - Minimum threshold: 5/10 (each)
- Final marking: weighted average provided all thresholds are met
- Second opportunity: only items below the threshold (or not delivered) in the first opportunity
- Policy of Zero-tolerance to plagiarism. We strongly encourage...
 - ...Proper citation of sources used
 - .. Authorship acknowledgement of the work by other

Calendar - Theory

General calendar of the course - 10,5 hours

1st bimester: 10/09 - 22/10

1. Introduction. (0.5h)
2. Problem solving in AI (3h)
3. Structured representations of knowledge (2h)
4. Methods of knowledge representation (3h)
5. Basic models of reasoning (2h)

Thursdays, 15:30 - 17:00

Calendar - Labs: Face2Face

General calendar of the course

1st bimester: 10/09 - 22/10

1. Solving problems by searching (3 sessions, 4.5 h)

Due date: 1 week after the last session of 1st part (8 Oct)

2. Ontologies (3 sessions, 4.5h)

Due date: 1 week after the last session of 2nd part (29 Oct)

UdC, UVIGO: Tuesday, 15:30-17:00

USC: Thursday, 17:00-18:30

Schedule

Expositive Lecture	Dates (Thu)	Interactive Session	Dates(Tue/Thu)
Intro + Problem Solving in IA	10 Sep		
Problem Solving in IA	17 Sep	Solving problems by searching	UDC, UVIGO: 15-SEP USC: 17-SEP
Problem Solving in IA	24 Sep	Solving problems by searching	UDC, UVIGO: 22-SEP USC: 24-SEP
Knowledge-Based Systems	1 Oct	Solving problems by searching	UDC, UVIGO: 29-SEP USC: 1-OCT
Methods of Knowledge Representation	8 Oct	Ontologies	UDC, UVIGO: 6-OCT USC: 8-OCT
Basic models of reasoning	15 Oct	Ontologies	UDC, UVIGO: 13-OCT USC: 15-OCT
Structured representations of knowledge	22 Oct	Ontologies	UDC, UVIGO: 20-OCT USC: 22-OCT

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A perspective on AI

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Past, Present and Why “Old” AI still Matters?

What is AI today?
(typical answer)



[Dartmouth Research Project on Artificial Intelligence \(1956\)](#)



[Dagstuhl Seminars](#)

Reasoning about brothers and sisters...

Julia has two sisters and one brother.

How many sisters does her brother
Martín have?



again

Sure! Let's go step by step again:

1. Julia has **two sisters** and **one brother**.

- So the children in the family are:
 - Julia
 - Julia's other sister
 - Julia's second sister? Actually, careful — "two sisters" from **Julia's perspective**. That means Julia sees **two other sisters**? Let's check.

We need to be precise:

- "Julia has two sisters" → Julia **plus two sisters**? No, usually it means **Julia herself is not counted**, so there are **two sisters besides Julia**.
- "and one brother" → that brother is Martín.

So the siblings are:

- Julia
- Her two sisters
- Martín

How many sisters does Martín have?

- Martín's sisters are **Julia + the two sisters of Julia** → **3 sisters**.

✓ **Correct answer: 3 sisters.**

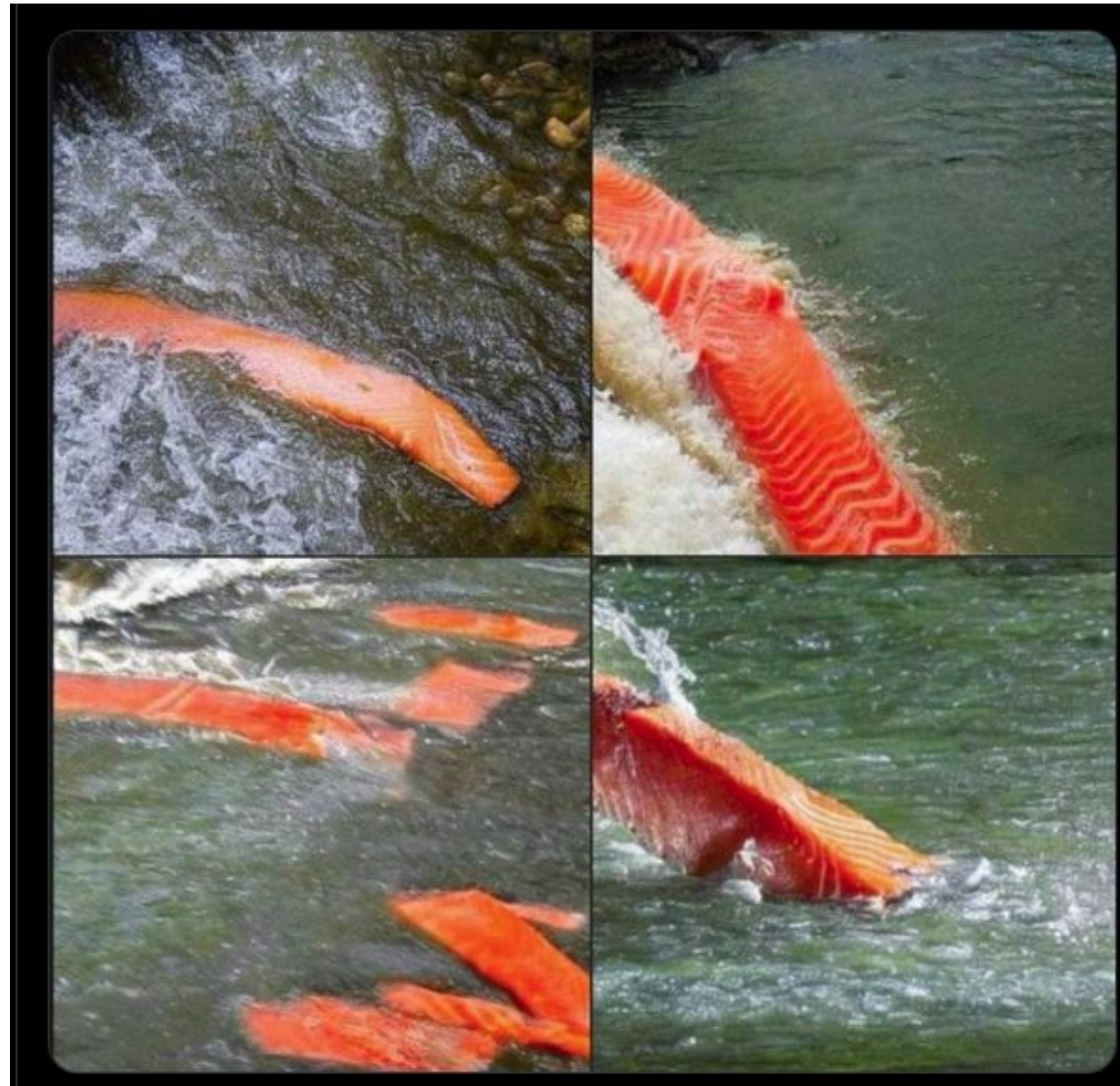
It's easy to miscount if you forget whether "Julia has two sisters" includes Julia or not.

Do you want me to draw a tiny diagram to make it crystal clear?

MARTÍN HAS TWO SISTERS.

The problem of the salmon in the river

“Draw a salmon in a river”





Past, Present and Why “Old” AI still Matters?

Few years ago reviewing a paper, I wrote:

25 "Do not boil the ocean. Is it really necessary to apply machine learning techniques to identify the data types (string, integer, float, date) of the input sources? I think that the definition of a **good set of heuristic-based rules** would be enough and better results will be obtained. In the related work is mentioned that these table interpretation approaches **use regular expressions to identify datatypes**, how are the results compared with the proposed solution of this paper?

The [paper](#) was rejected in that conference, but the authors won the best paper award in the next one

Take home message (from the whole subject)

 **Search** → Still the foundation for decision-making and planning.

 **Knowledge-Based Systems** → Provide explainability and reliability.

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 **Ontologies** → Structure and integrate knowledge across domains. Reasoning!

 **Past ≠ Obsolete** → “Old” AI techniques are building blocks for today’s systems.

 **Don’t reinvent the wheel** → Learn from classical AI to design (better) solutions.



A perspective on AI

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